

Analysis of Surrogate Indicators for Evidence of Subsurface Preferential Flow Pathways:

Impact of subsurface preferential flow on variability of NDVI

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Abstract— Watershed-scale processes governing chemical fluxes to adjacent ecosystems are so poorly understood that effective strategies for mitigating chemical contamination cannot be formulated. Characterization of hydrologic processes and chemical behavior at the watershed scale is critical to the development of sustainable agricultural practices. Identifying locations for monitoring hydrologic processes like subsurface preferential flow is difficult because conventional sampling methods are inadequate for measuring this highly variable, yet critical process. A method for detecting and mapping subsurface preferential flow pathways based primarily on ground-penetrating radar (GPR) data and digital elevation maps (DEM) was developed. This procedure was confirmed for a Maryland cornfield using real-time soil moisture data, maps of within-field grain yield, and remotely sensed imagery. Unfortunately, it is economically unfeasible and logistically impractical for producers to use the GPR-DEM procedures to map subsurface preferential flow pathways for all crop fields. It may, however, be possible to use remotely sensed imagery, grain yield maps, and a DEM as surrogate indicators of subsurface preferential flow pathways occurring at or near crop rooting depth. The normalized difference vegetation index (NDVI) shows an increase with distance from primary, secondary and tertiary preferential flow pathways during above-normal rainfall growing seasons. There appears to be a decrease of NDVI with distance from preferential flow pathways for a dry/drought year. Imagery collected during drought conditions appears especially useful as only within-field locations with subsurface irrigation from preferential flow pathways maintain vigor. Maps of vegetative productivity derived from remotely sensed imagery may be more useful than within-field grain yield maps for detecting and delineating locations of subsurface preferential flow pathways. The ability to delineate field locations with a high probability of subsurface preferential flow pathways will allow producers to better manage crop production, and mitigate losses of agricultural chemical inputs to neighboring ecosystems and waterways.

Keywords—subsurface preferential flow pathways, soil moisture, NDVI, airborne imagery

I. INTRODUCTION

Watershed-scale processes governing chemical fluxes to adjacent ecosystems are so poorly understood that effective strategies for mitigating chemical contamination cannot be formulated. Characterization of hydrologic processes and chemical behavior at the watershed scale is critical to the development of sustainable agricultural practices. Identifying locations for monitoring hydrologic processes like subsurface preferential flow is especially difficult because conventional sampling methods are inadequate for measuring this highly variable, yet critical process. A method for detecting and mapping subsurface preferential flow pathways based on ground-penetrating radar (GPR) data and digital elevation maps (DEM) was validated for a cornfield in Maryland using real-time soil moisture data, maps of within-field grain yield, and remotely sensed imagery.

Unfortunately, it is unfeasible and impractical for producers to use the GPR-DEM procedures to map subsurface preferential flow pathways for all fields. It may, however, be possible to use remotely sensed imagery, grain yield maps, and a DEM as surrogate indicators of subsurface preferential flow pathways occurring at or near crop rooting depth. Prior analysis of corn yields showed a decrease with increasing distance from GPR-DEM identified preferential flow pathways during a drought year [1]. However, within-field indications of soil moisture variability may be subdued in grain yield data by the ability of the crop to adapt to soil moisture stress during grain-filling. Resources from leaves and stalks are translocated to grain by plants to compensate for limited moisture availability; whereas, foliage elements suffer a loss of productivity. It is therefore suggested that crop foliage may be a more sensitive indicator of soil moisture conditions.

If foliage is more sensitive to soil moisture conditions than grain yield, then growing seasons experiencing extreme moisture conditions (i.e. precipitation deficit/drought versus precipitation excess/flooding) would be expected to hold the greatest information. Growing seasons with limited rainfall would have decreasing yield with increasing distance from

flow pathways occurring at or near crop rooting depth. Only plants occurring near subsurface flow pathways would receive sufficient moisture to produce. Conversely, growing seasons with abundant or heavy precipitation would tend to have increasing yield with increasing distance from subsurface flow pathways because of excessive water availability.

The objective of the analysis reported here was to investigate the impact of subsurface flow pathways on imagery of crop foliage. The relationship between a spectral vegetation index and distance from subsurface preferential flow pathways was investigated using imagery collected during wet and dry growing seasons.

II. METHODS

A. Study Site

The research site is a 22 ha agricultural field located at the USDA, Beltsville Agricultural Research Center, Beltsville, Maryland. The site is part of the Optimizing Production inputs for Economic and Environmental Enhancement (OPE3) study. OPE3 seeks to compare agricultural production systems at a scale large enough to capture the spatial variability of crop and soil parameters, yet small enough for fields to be in similar climatic and geologic settings (<http://hydrolab.arsusda.gov/ope3/>).

The site contains four small, bounded watersheds, from 3.2 to 4.2 ha, feeding a wooded riparian wetland and first-order stream. A DEM developed from kinematic global positioning system (GPS) data acquired on a 5 m grid showed that 74% of the site has a slope < 2% and only 2% of the site has a slope > 3%. The soils are sandy textured with buried clay lens (coarse-loamy, siliceous, semiactive, mesic, Typic Hapludult).

B. Delineation of Preferential Flow Pathways

Subsurface preferential flow pathways were delineated using GPR data and the DEM [2]. A subsurface interface radar (SIR) was calibrated on two locations representing the range of soil profiles of the experimental site. A 150 MHz frequency GPR antenna was dragged across the soil surface and the return times of the GPR signals from steel calibration plates were used to calculate soil dielectric constants. GPR data were acquired for the 22 ha site at two scales of observation and a digital trace of the subsurface reflections was produced. The first GPR data consisted of parallel North-South transects, 25 m apart over the entire watershed. The second GPR data set consisted of parallel North-South transects, 2 m apart over eleven 25 x 25 m

plots within the small watershed. The spatial autocorrelation of the 40 km of subsurface reflections were determined and then the depth of subsurface reflections was calculated for 8 m x 8 m cells for the entire watershed. The depth to the GPR-identified clay lens was subtracted from the DEM to determine an elevation-corrected map of the subsurface stratigraphy. Geographic Information System (GIS) hydrologic modeling tools FLOW DIRECTION and FLOW ACCUMULATION were applied to the elevation-corrected subsurface topography to determine potential flow pathways [3]. The FLOW DIRECTION command was used to create a grid indicating the direction of flow from each cell to its steepest down-slope neighboring cell and the FLOW ACCUMULATION command was then applied to the FLOW DIRECTION grid to create a grid of accumulated water flow from its surrounding cells for a given mass of applied water. At least 21, 8, or 2 cells must drain into a pathway to be considered a primary, secondary, or tertiary path, respectively.

C. Remotely Sensed Data

Color infrared aerial photography was collected August 15, 1999 and digitized to produce separate images for the green, red and near infrared (NIR) emulsion layers at 0.5 m resolution. Calibrated hyperspectral imagery of the study site was collected with the airborne imaging spectrometer for applications (AISA) sensor July 18, 2000 (1.0 m resolution), and August 22, 2001 (2.5 m resolution). Apparent surface reflectance was calculated using coincident downwelling spectral radiance acquired with an upward-looking fiber optics probe fed into the AISA detector array. The imagery from all three years was re-sampled to 8 m x 8 m pixels. The normalized difference vegetation index (NDVI) was calculated from:

$$NDVI = \frac{R_{NIR} - R_{red}}{R_{NIR} + R_{red}} \quad (1)$$

where R_{NIR} and R_{red} are the measurements of reflected NIR and red radiance from the surface. This corresponds to digital numbers (DN) of the digitized emulsion layers of the 1999 aerial photograph, and percent apparent reflectance of the 845 nm (NIR) band and 680 nm (red) from the 2000 and 869 nm (NIR) and 690 nm (red) 2001 AISA

Manufacturer names are for information purposes only and do not imply endorsement by USDA to the exclusion of others.

imagery. The NDVI has been related to fraction of absorbed photosynthetically active radiation, foliage density expressed as leaf area index (LAI), percent vegetative cover, and biomass, and is thus, a useful index of crop productivity or cumulative growth. The NDVI from zones with distances 0-5m, 5-10m and 10-15m from the GPR/DEM-identified primary, secondary and tertiary flow pathways were extracted from the imagery using GIS methods. The means of these NDVI buffer zones were calculated and plotted against distance from flow pathway locations. The number of pixels for each buffer zone are listed as Table 1. Differences between the 1999 digitized aerial photograph and 2000 and 2001 AISA data pixel counts are due to slight differences of coverage resulting from manual determination of the extent of planted fields.

A drought during the 1999 growing season which severely reduced vegetative growth and yield (i.e. a “dry” year). Precipitation was frequent and of moderate intensity during the 2000 growing season which produced vigorous vegetative development and high yields (i.e. a “wet” year). During 2001 precipitation was adequate for vegetative growth and average yields, although less than that of the 2000 growing season (i.e. an “average” year). Thus, the imagery captures vegetative productivity for growing seasons experiencing dry, wet, and average precipitation [4].

III. RESULTS

Plots of NDVI versus distance from primary, secondary, tertiary, and mean for all three flow pathways are shown by Figs 1 through 3. The 1999 data (“dry year”) show a decrease with distance from primary to secondary, and tertiary flow pathways (Fig. 1). The tertiary flow pathway shows the most dramatic NDVI decrease with distance. The mean of all three NDVI buffer zones also shows a slight decrease with distance and is the same as the secondary flow pathways. The 2000 data (“wet year”) show the opposite trend of the 1999 data with an increase of NDVI with distance from flow pathway. The mean values also follow this trend. The 2001 data (“average” year) show a trend similar to the 2000 data for the 0-5m and 5-10m buffer zones followed by a decrease by the 10-15m buffer zone. The trend is not as strong as that of the 2000 data. Although the NDVI changes with distance are modest for all three years, the large number of pixels used to examine the changes suggests that the trends are present.

IV. CONCLUSIONS

Increasing distance from flow pathways near crop rooting depth during dry years results in decreased crop productivity. Increasing distance from flow pathways during years of

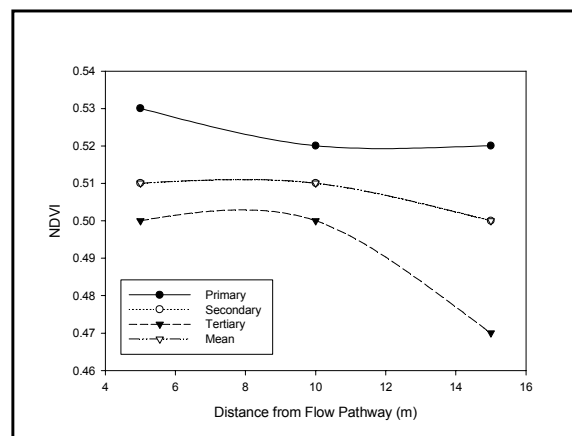


Figure 1. Dry year NDVI derived from 1999 digitized color infrared photography versus distance from primary, secondary, tertiary, and mean of all three subsurface preferential flow pathways.

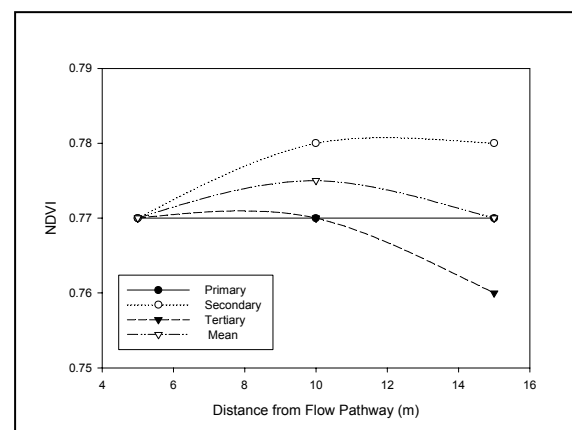


Figure 2. Average year NDVI from 2000 AISA imagery versus distance from primary, secondary, tertiary, and mean of all three subsurface preferential flow pathways

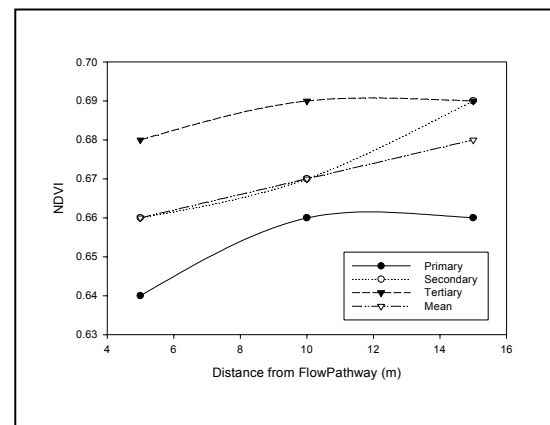


Figure 3. Wet year NDVI derived from 2001 AISA imagery versus distance from primary, secondary, tertiary, and mean of all three subsurface preferential flow pathways

adequate or excessive rainfall results in increased crop productivity. During dry years subsurface flow pathways irrigate the crop while the rest of the field relies solely on water

from soil moisture fed by rainfall. As distance from flow pathway increases, there is less water available to the crop and thus, the effects of drought on productivity increase. During wet years, locations close to flow pathways receive water via rainfall and via the flow pathways. This results in decreased productivity closer to flow pathways from saturation, especially during growing seasons of excessive rainfall.

The results indicate that multispectral imagery processed to NDVI provides indications of the presence of subsurface preferential flow pathways. The dry/drought year data appear to be especially useful for revealing spatial relationships between vegetation productivity and soil moisture patterns. A *caveat* of the analysis is that surface topography and soil texture may also produce similar patterns. The degree to which landscape curvature is correlated with grain yield and above-ground plant productivity as captured by NDVI needs investigation. The ability to infer moisture characteristics via analysis of imagery of overlying vegetation holds promise [4,5]. These results when combined with the information from grain yield maps and topographic curvature analysis may permit the stratification of a crop field into zones of probability-of-occurrence for subsurface preferential flow pathways. The ability to delineate within-field locations with a high probability of subsurface preferential flow pathways will allow producers to better manage crop production, and mitigate losses of agricultural chemical inputs to neighboring ecosystems and waterways. .

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TABLE I. Buffer Zone Pixel Counts

	Primary			Secondary			Tertiary		
	Buffer Zone (m)								
Year	0-5	5-10	10-15	0-5	5-10	10-15	0-5	5-10	10-15
1999	493	215	548	1039	215	930	2120	579	624
2000	301	566	247	656	1085	443	1508	1460	353
2001	301	566	247	656	1085	443	1508	1460	353